AMCS1034

Software Development Fundamentals

Assignment Submission Form

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Submission Date	20 December 2025

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Report DCS1S2G2 Pet Care Assistant App.docx

REPORT CREATED
Dec 20, 2025

Summary

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🐾 System Description:

The software application that we had chosen for our software development project is the Pet Care Assistance app.The sub applications we had developed are feeding tracker,vet appointment log and grooming schedule. This Pet Care Assistant App is designed to assist the pet owners manage daily care routines and important events for their loving pets. It provides a centralized platform to track feeding times, vet appointments, and grooming schedules for their pets . The system ensures structured and timely pet care, reducing the risk of missed tasks and improving overall pet health care .

Objectives

- Simplify pet care management for busy owners with tkinter and customtkinter advanced GUI design.
- Provide logs in each system for essential activities with tkcalender.
- Maintain records of pet health and grooming history by using json and txt file.
- Enhance organization and reduce stress in pet care routines.

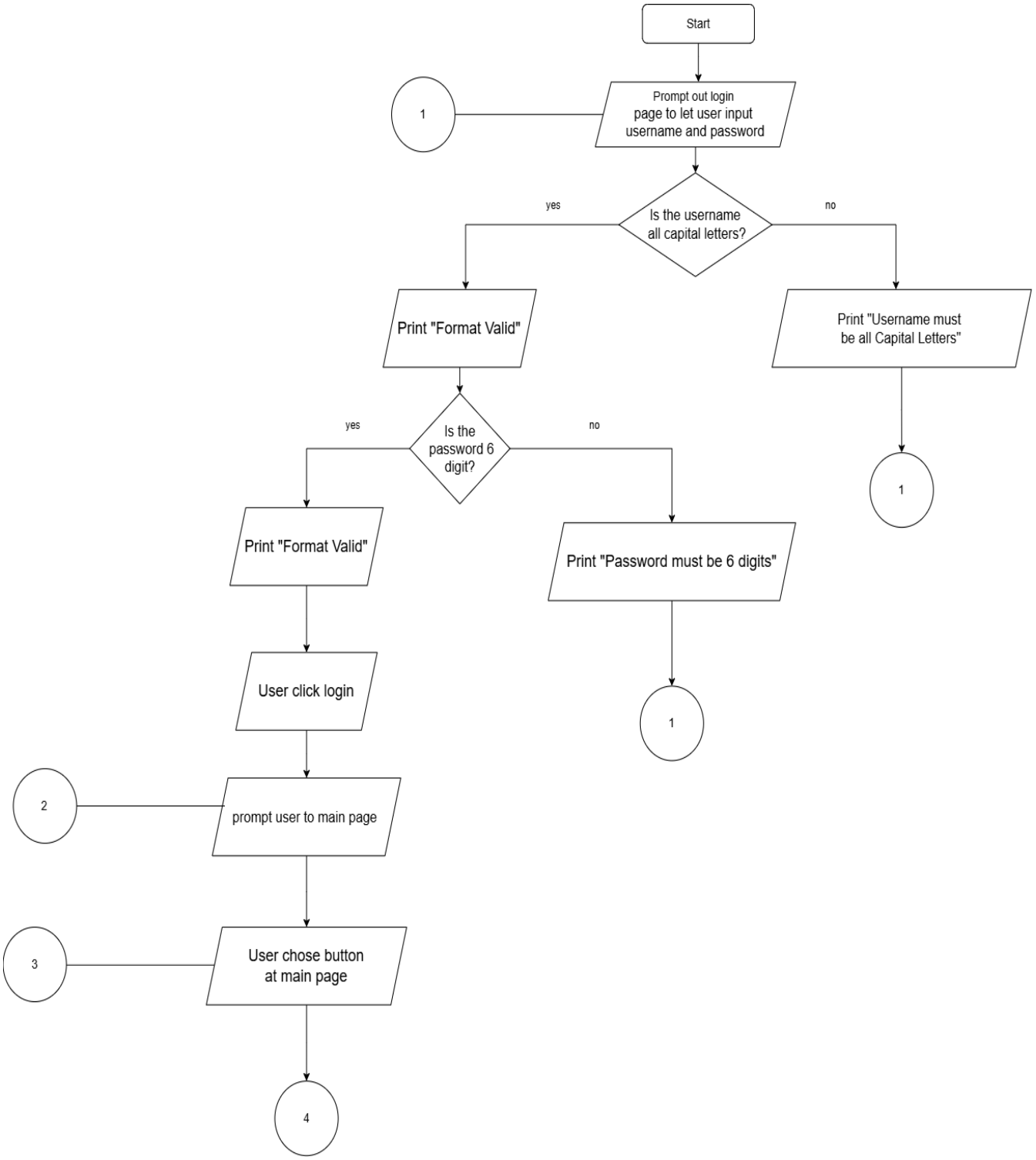
Key Features

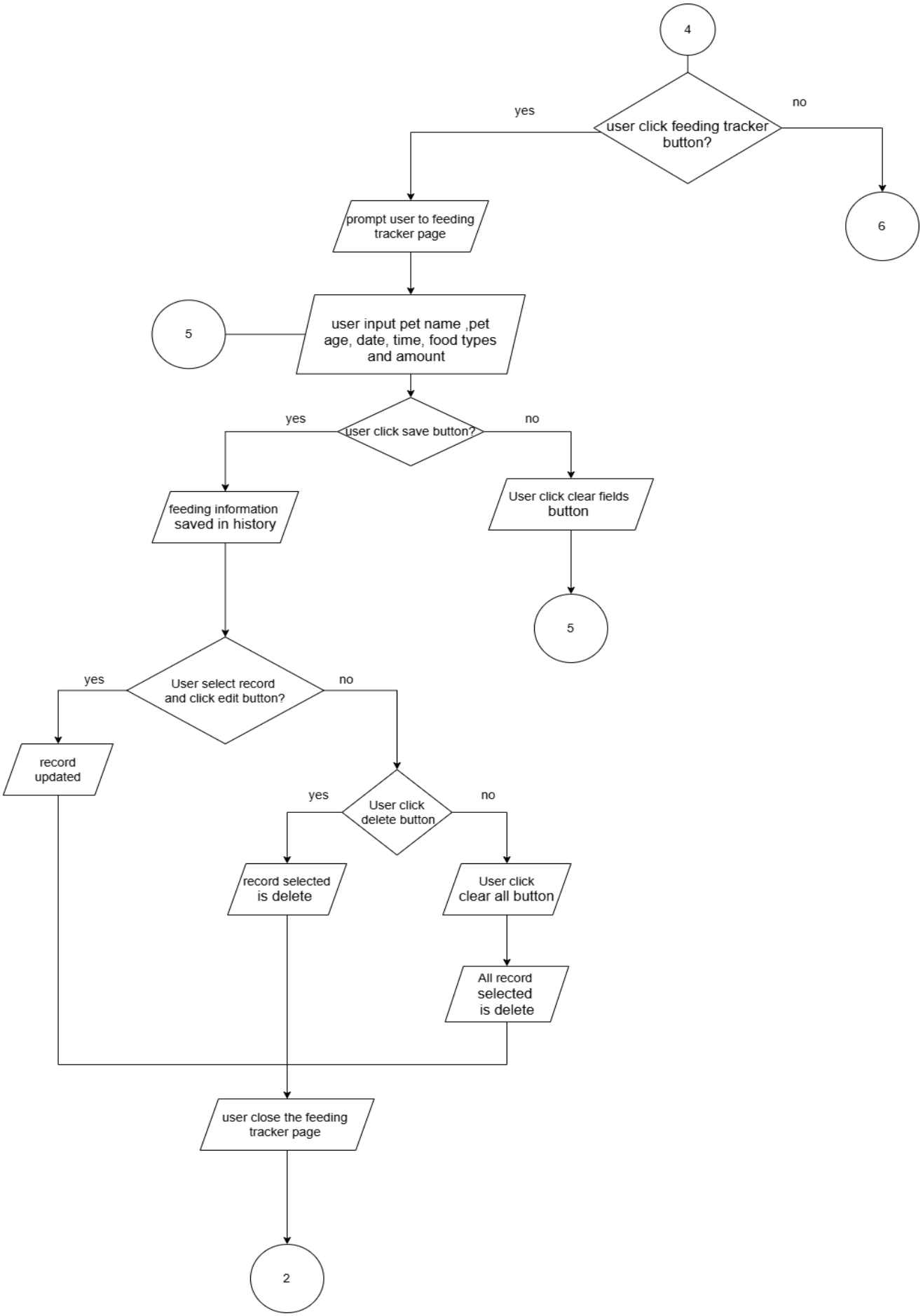
- Feeding Tracker
  - Log of meals with time and type of food.
  - Schedule feeding times.
- Vet Appointment Log
  - Record upcoming vet visits.
  - Track completed vet visit.
- Grooming Schedule
  - Plan grooming tasks (bath and blow dry, nail clipping,brushing, ear care).
  - Set recurring schedules.
  - Track completed grooming activities.

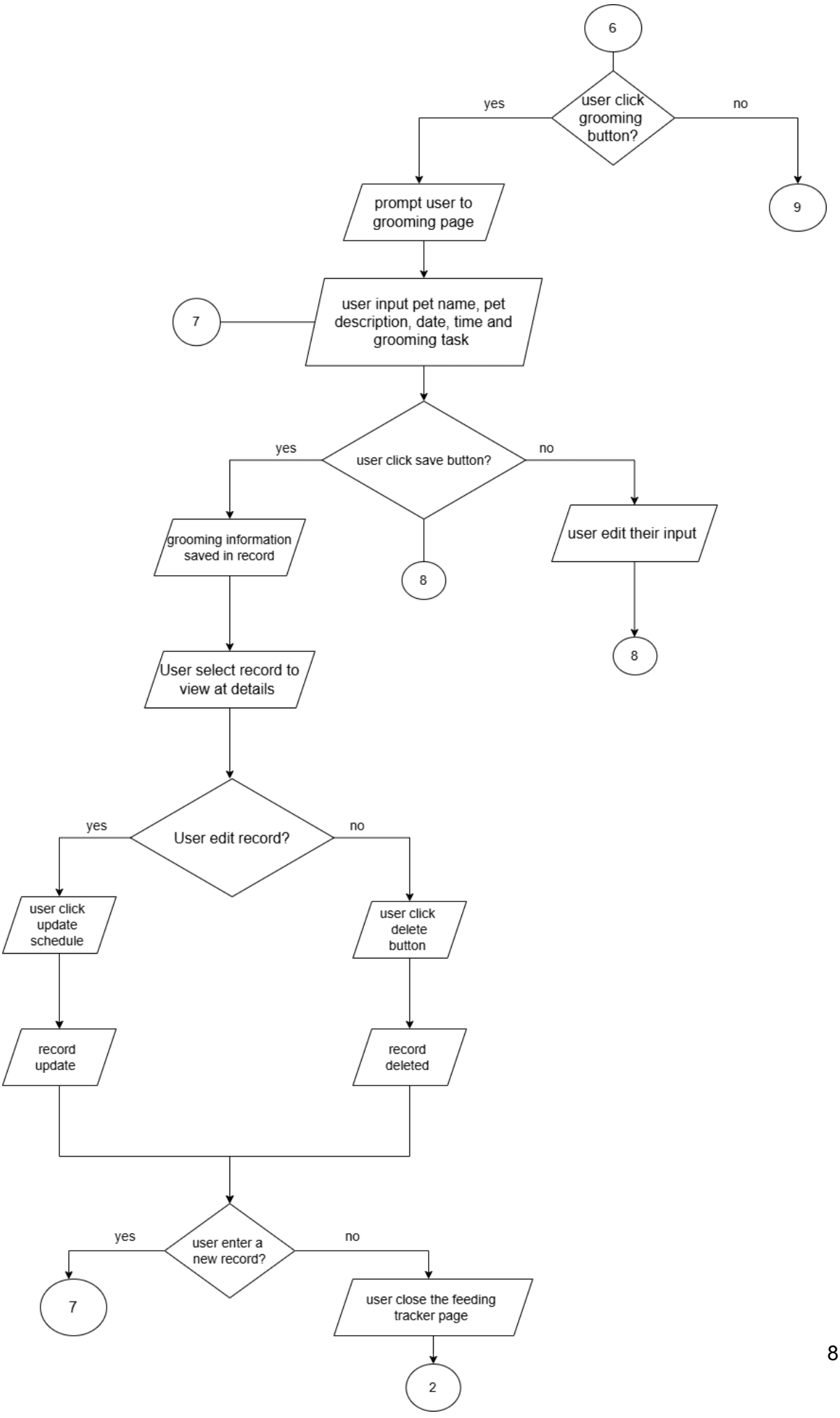
System Benefits

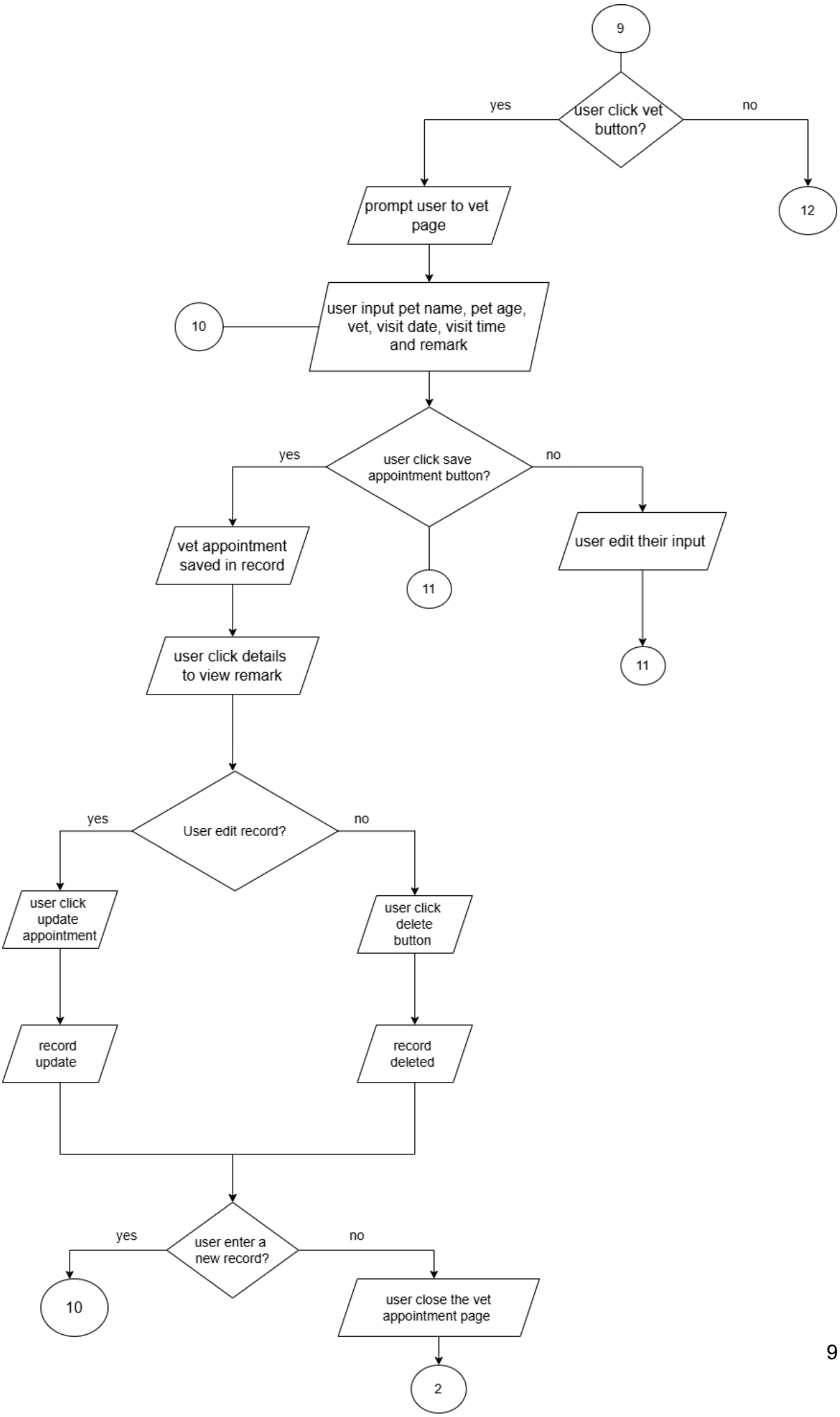
- Provides a tidy and clear record platform.
- Provides a platform for all pet-related records.
- Saves time and reduces the chance of forgetting important tasks.
- Ensure user discipline in pet care.

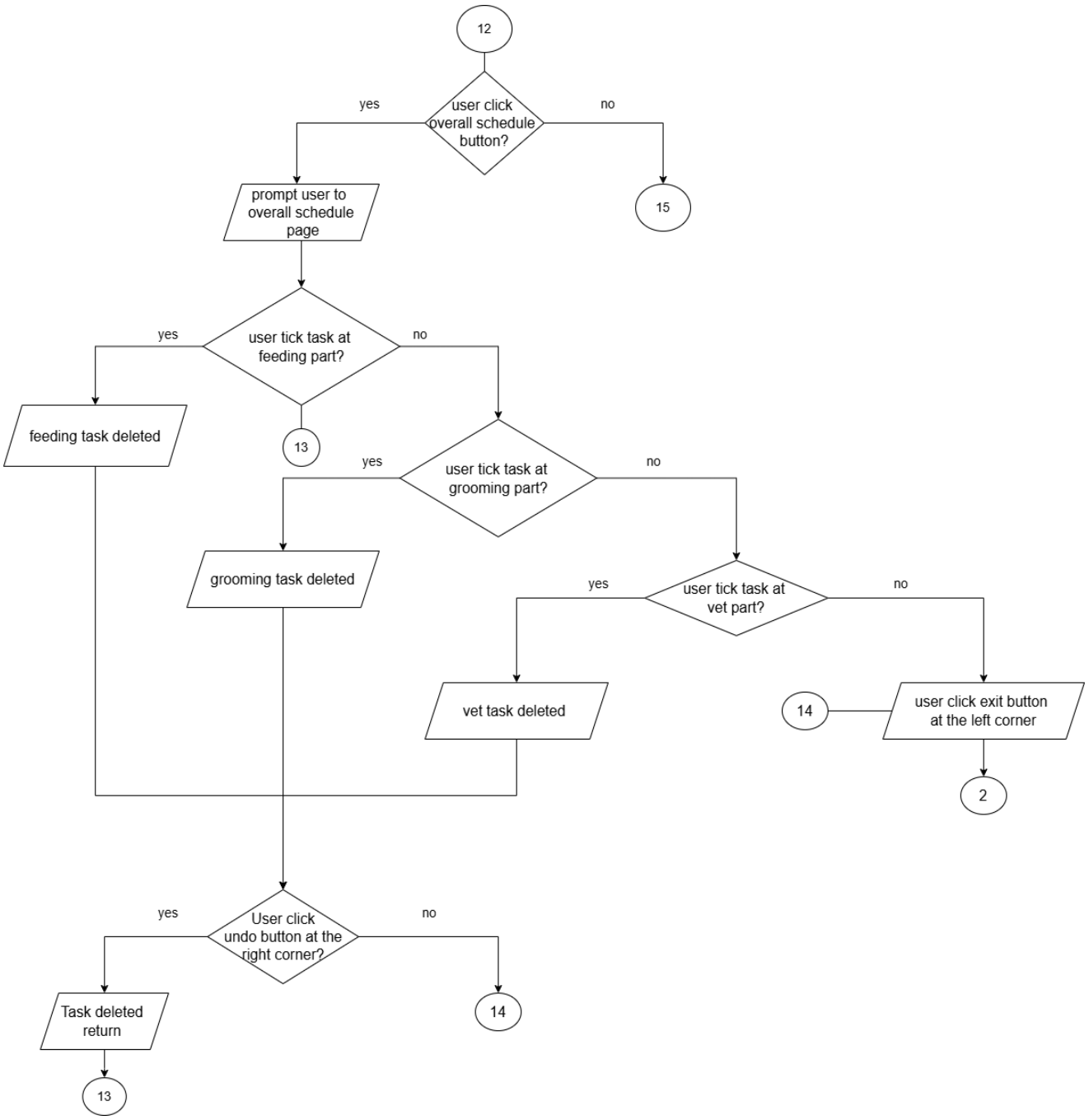
Stepwise Refinement

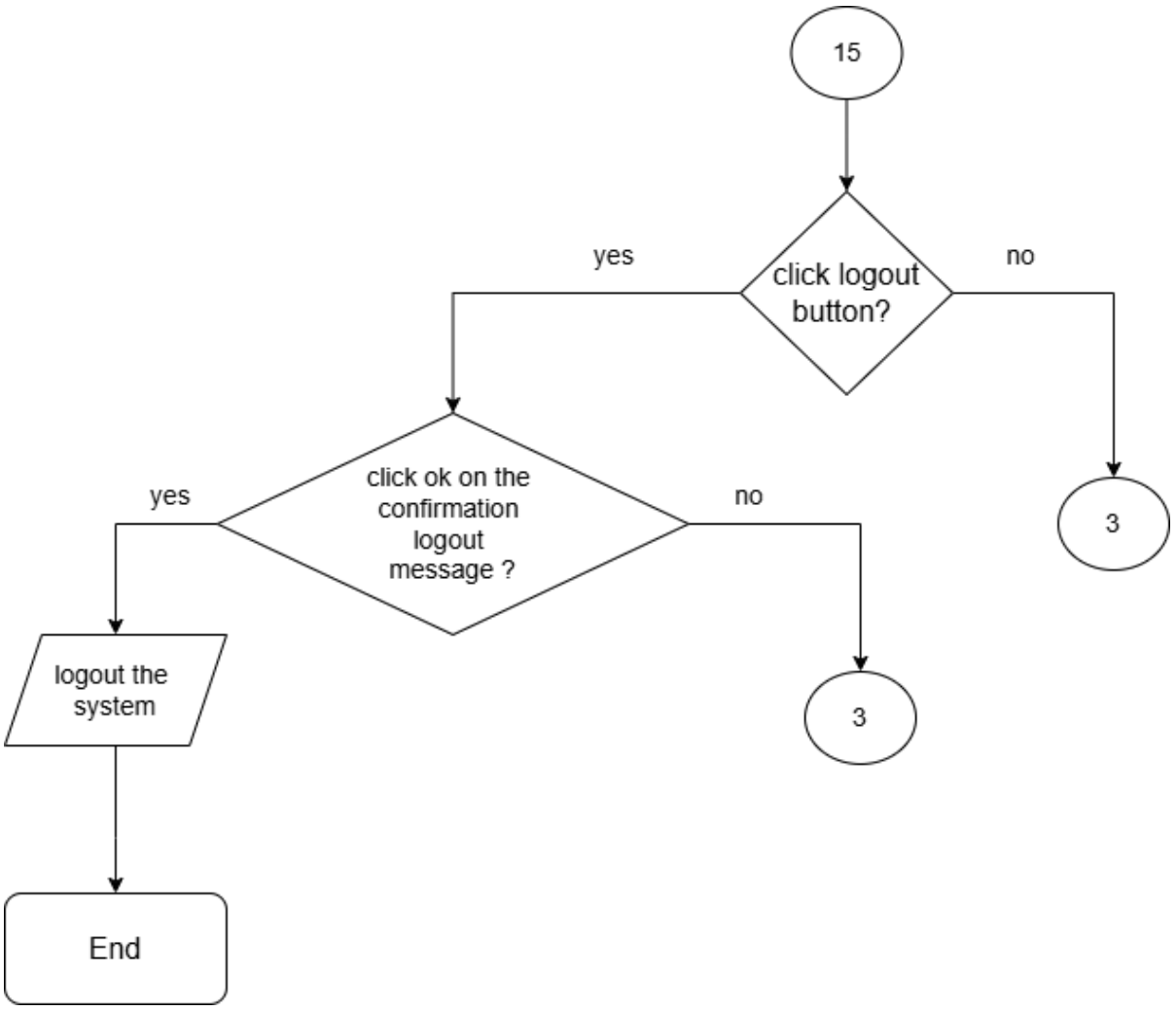



4
user click feeding tracker button?
yes
no
6
prompt user to feeding tracker page
5
user input pet name ,pet age, date, time, food types and amount
user click save button?
yes
no
feeding information saved in history
User click clear fields button
5
User select record and click edit button?
yes
no
record updated
User click delete button
yes
no
record selected is delete
User click clear all button
All record selected is delete
user close the feeding tracker page
2


6
user click grooming button?
yes
no
prompt user to grooming page
9
7
user input pet name, pet description, date, time and grooming task
user click save button?
yes
no
grooming information saved in record
8
user edit their input
8
User select record to view at details
User edit record?
yes
no
user click update schedule
user click delete button
record update
record deleted
user enter a new record?
yes
no
7
user close the feeding tracker page
2

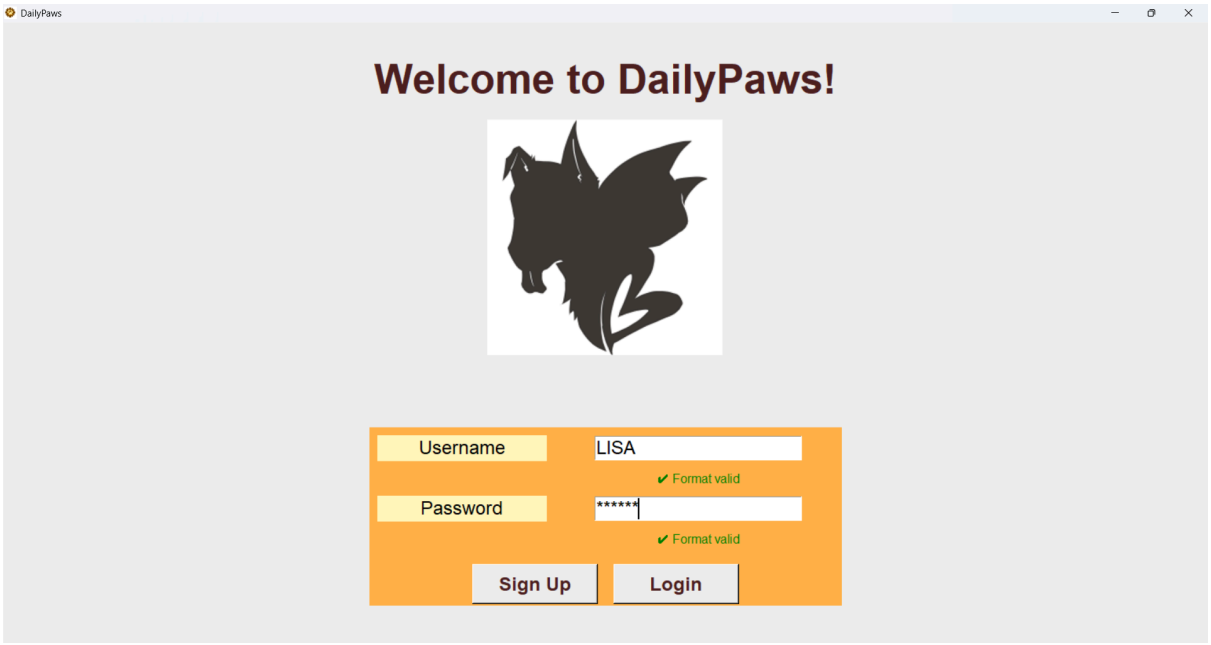

9
user click vet button?
yes
no
prompt user to vet page
12
10
user input pet name, pet age, vet, visit date, visit time and remark
user click save appointment button?
yes
no
vet appointment saved in record
11
user edit their input
11
user click details to view remark
User edit record?
yes
no
user click update appointment
user click delete button
record update
record deleted
user enter a new record?
yes
no
10
user close the vet appointment page
2


12
user click overall schedule button?
yes
no
prompt user to overall schedule page
15
user tick task at feeding part?
yes
no
feeding task deleted
13
user tick task at grooming part?
yes
no
grooming task deleted
user tick task at vet part?
yes
no
vet task deleted
14
user click exit button at the left corner
2
User click undo button at the right corner?
yes
no
Task deleted return
13
14


15
click logout button?
yes
no
3
click ok on the confirmation logout message ?
yes
no
3
logout the system
End

🐾User Manual

1.login page



a)User must login with their username in all capital letter

-If user enter wrong input that is no require the system requirement which is username must be all capital letter,the system will prompt a warning line at the bottom of the username input box said “username must be ALL CAPITAL LETTER"

-If user enter correct input ,the system will prompt a verification line at the bottom of the username input box said “Format valid”

b)User must login with their password in 6 digit

-If user enter wrong input that is no require the system requirement which is password must be all capital letter,the system will prompt a warning line at the bottom of the username input box said “password must be 6 digits”

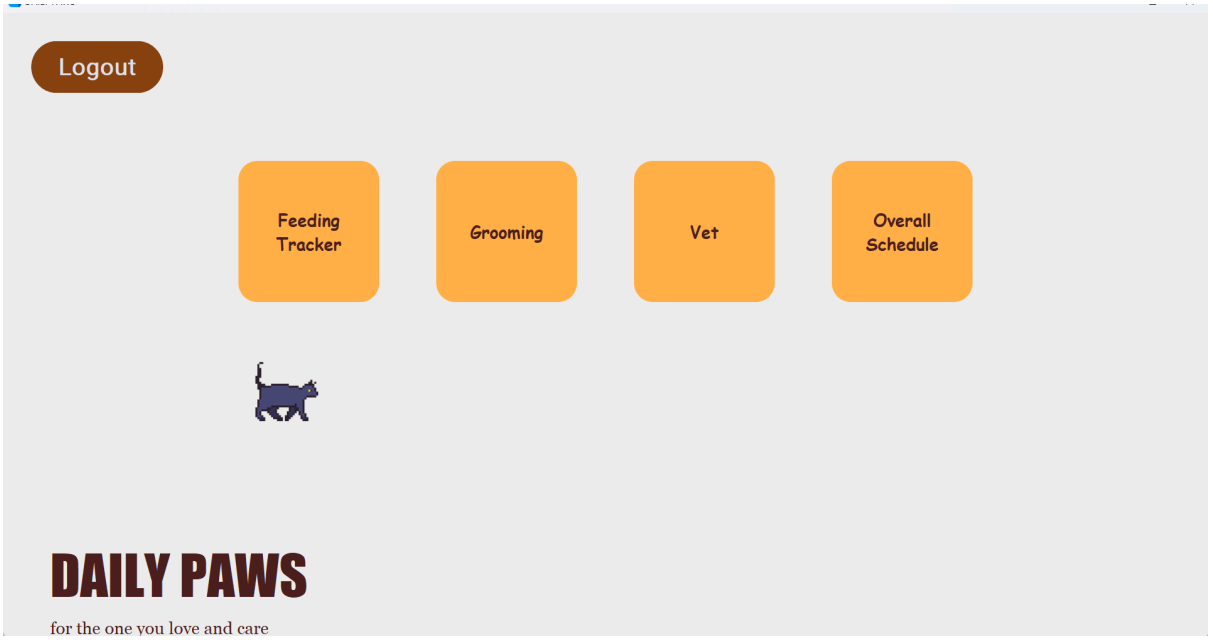
-If user enter correct input ,the system will prompt a verification line at the bottom of the username input box said “Format valid ”

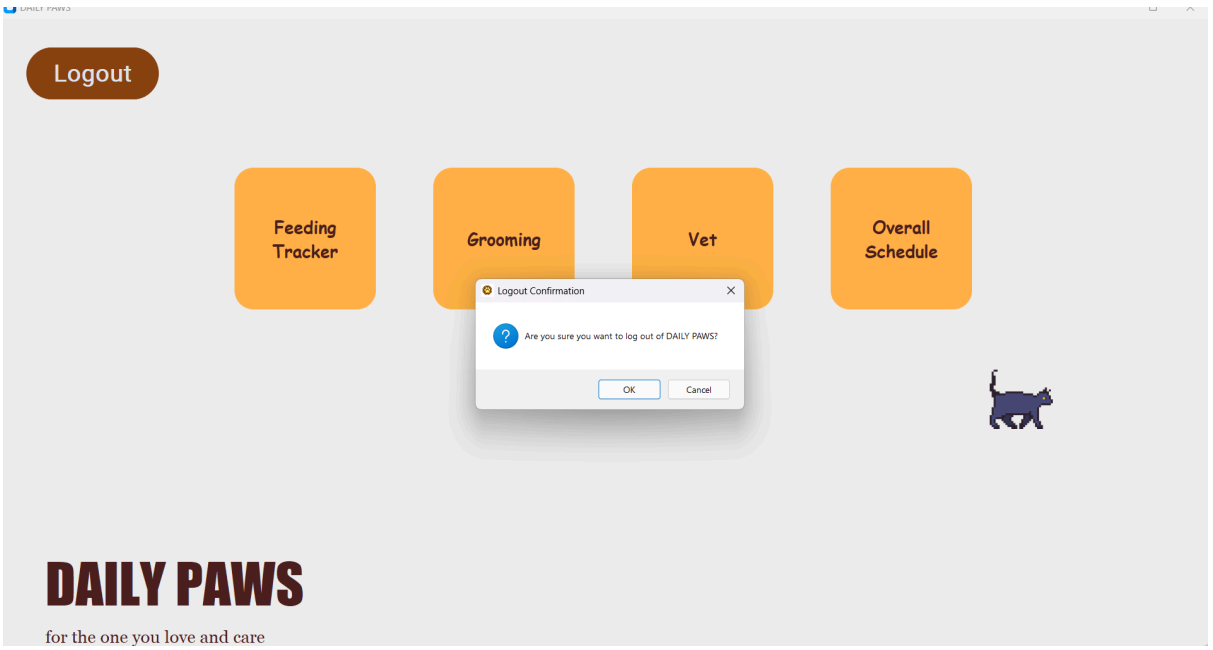
c)Sign in new username and password

-If a user inputs another new username and password they cannot log in.So, the user has to press sign in and click login again when shows sign in successfully.

- If a user uses the same username , the system will ask the user to try another username or the user can try to login if the username belongs to his or hers.

2.Main Page





a)Four main function button in the middle part of the main page

-user can click the Feeding Tracker button to access the Feeding Tracker page.

-user can click the Grooming button to access the Grooming page.

-user can click the Vet button to access the Vet Appointment page.

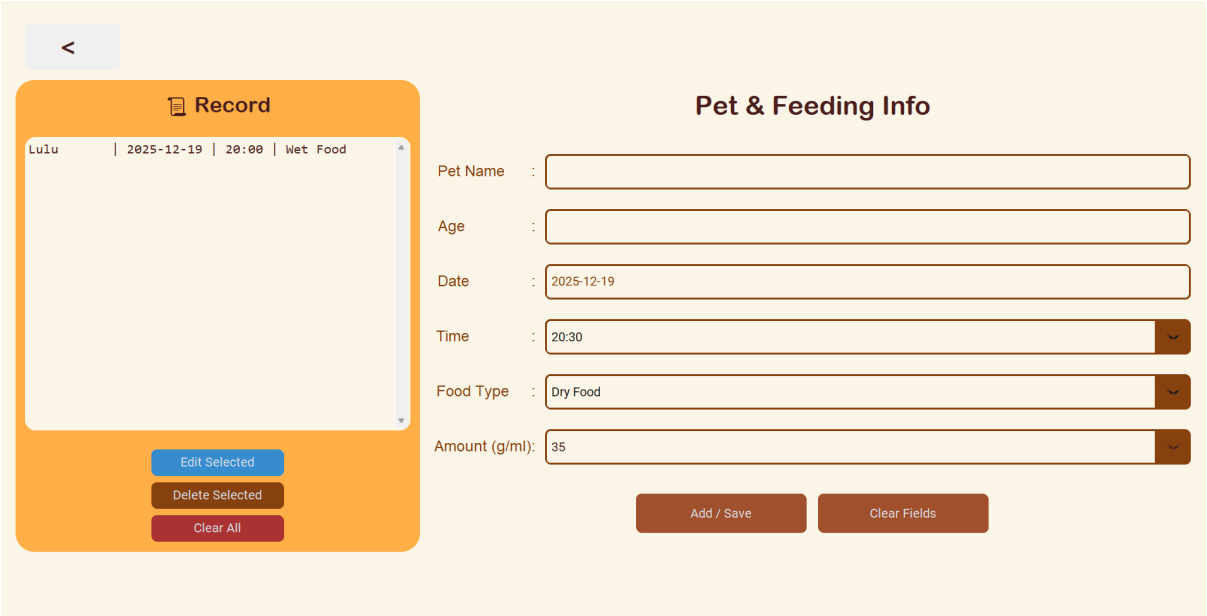
-user can click the Overall Schedule to access the overall schedule page.

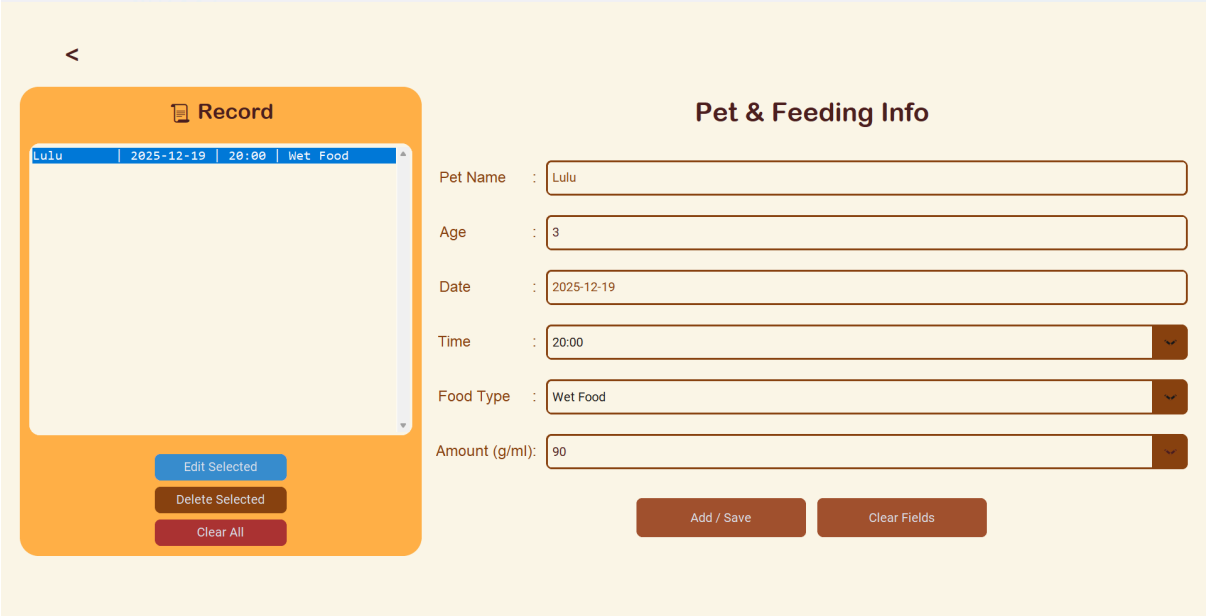
b)Logout Button

-When the user clicks the logout button located on the left corner of the main page , it will prompt a confirmation message for the user to choose.

-user can log out the whole system when the user clicks ok, if the user clicks the right corner cancel icon of the confirmation message box , it will remain on the main page.

3.Feeding Tracker





a)Insert Pet & Feeding Info

-Users can input the pet's information and the feeding info on the right which include pet name, pet age, the date and time you feed your pet, the food type that you feed and also the amount of food you have fed.

-For the pet age, only 2 digits can be inserted.

- The Date and time can be selected by clicking the input bar.

-The food type and amount are able to be selected by clicking the brown button at right of their input bar. The food type is limited for dry food, wet food, snacks ,milk and water. The limited range of amount is 35-110.

b)Save Record
-After fully filling up the information, users can click on the add/save button and the information will show in the record.

c)Delete Record
-Users can click on the record they wanted to delete and click the ‘delete selected’ button to delete it.

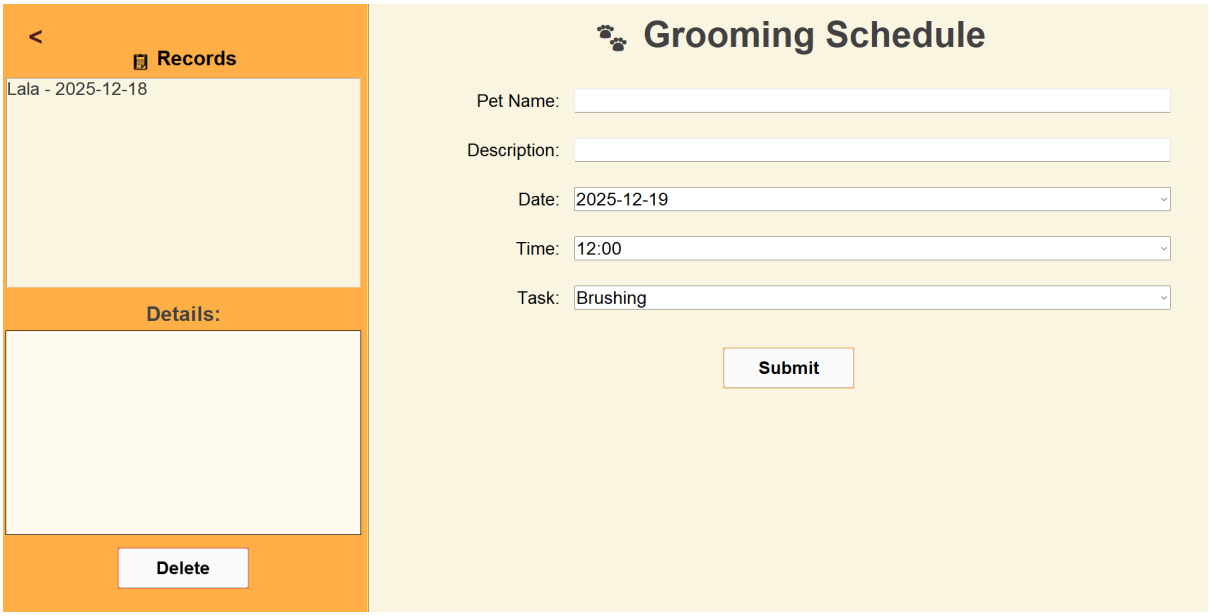
d)Edit Record
-Users can click on the record they wanted to edit and click the ‘edit selected’ button, the pet and feeding info will show the record you selected and you can edit it then click on the save button. The history will replace the record with the record that you have edited.

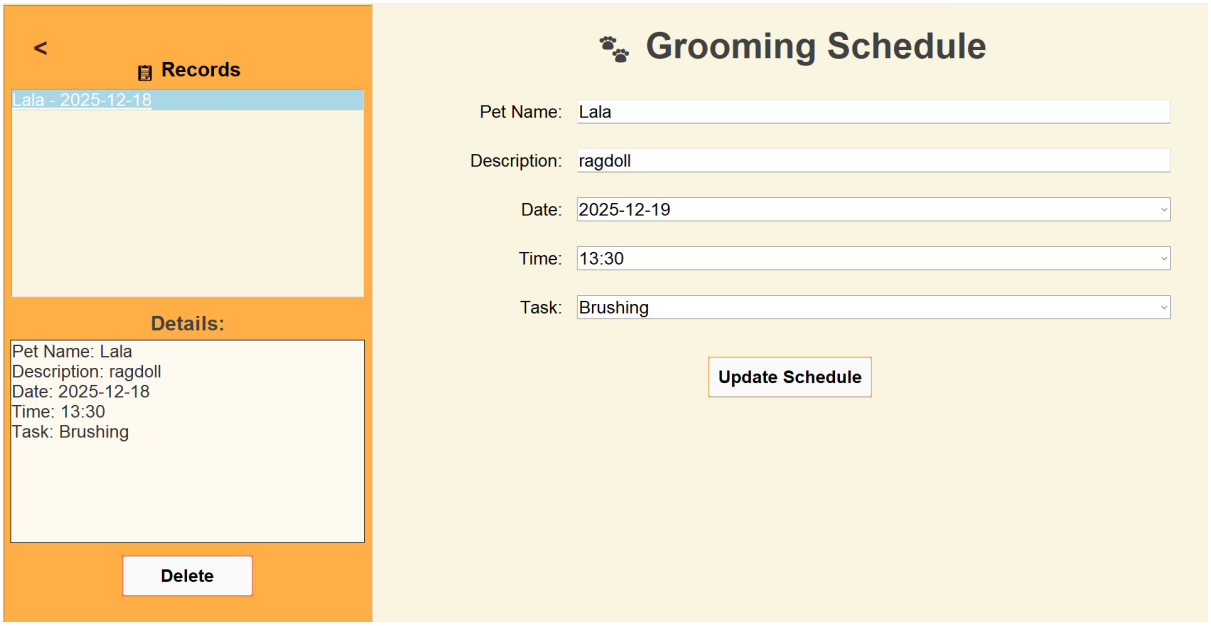
e)Clear Fields
-Users can clear the info that they selected and renew the time through clicking the ‘clear fields’ button.

f) Clear All
-Users can delete all the records in history through clicking the clear all button.

g)Back Button
-Users can click on the “<” button on the left corner to go back to the mainpage.

4.Grooming





a)Input Data

-Users can input their grooming data on the right which contain pet name, pet description and grooming task.

-Users need to select the date and time for their pet grooming task by clicking the button at the right end of its input bar.

-for the task is also can be selected by clicking the the button at the right end of its input bar. They can choose task like bath and blow dry ,nail clipping ,brushing and ear care.

b)Submit Data
After that, users can submit the data they have inserted and the data will appear on the left part which is the records part, it will show the pet name and date for grooming.

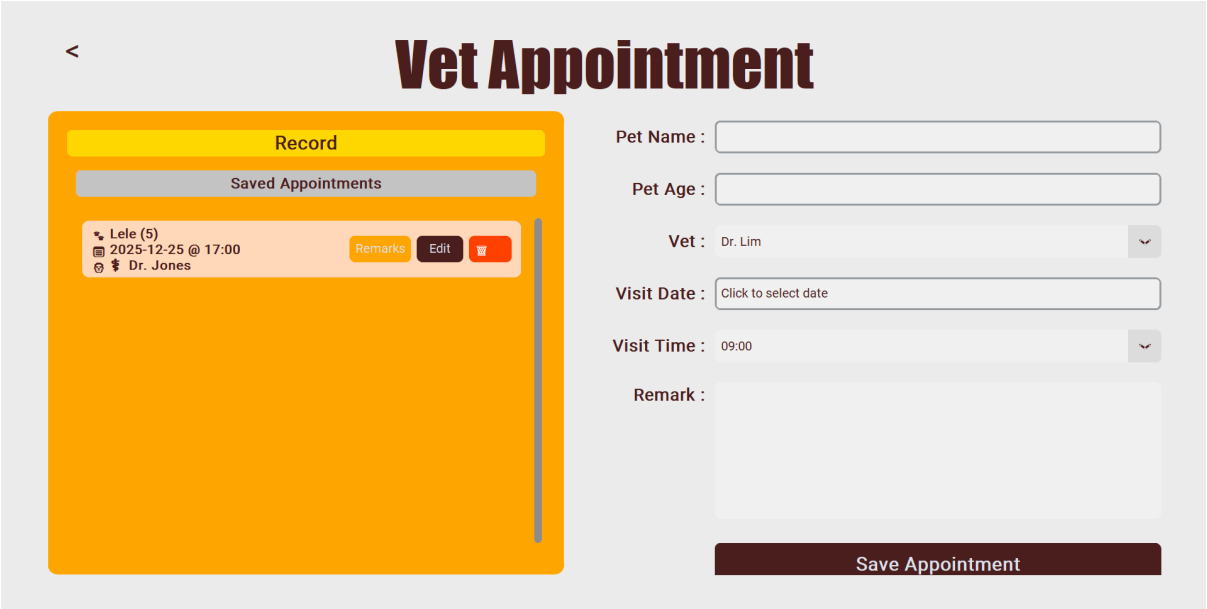
c)Details
User can also click the records to see the details which located below the record part

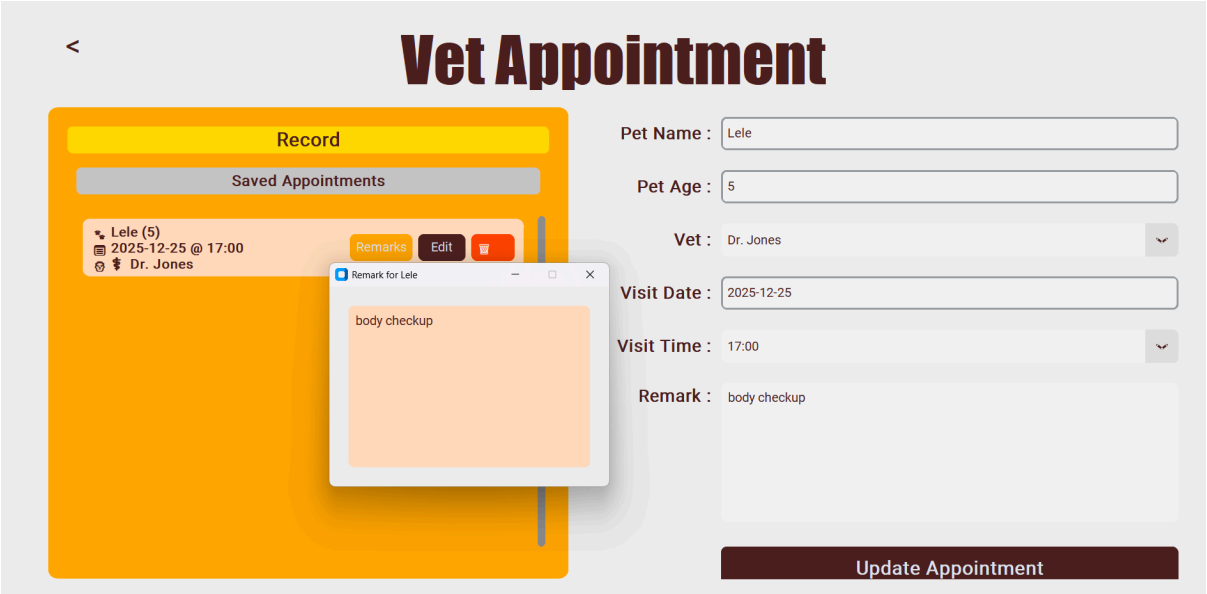
d)Update record
-User can also click the records to update, user can click the record that they want to edit and edit the input on the right part of the page to update schedule

e) Delete the record
-Users can also select the record that they want to delete and click the delete button to delete.

f)Back Button
-Users can click on the “<” button on the left corner to go back to the mainpage.

5.Vet appointment





a)Input Data

-Users can input their vet appointment data on the right which contain pet name, pet age ,vet,visit date, visit time and remarks.

-For the pet age, only 2 digits can be inserted.

-The vet, visit date and visit is limited.

b)Must Procedure to Save Data

-Users must insert all parts at the right to proceed to save appointments .

c)Record
-After that, users can submit the data they have inserted and the data will appear on the left part which is the records part, it will show the pet name and date and time and the vet visit.

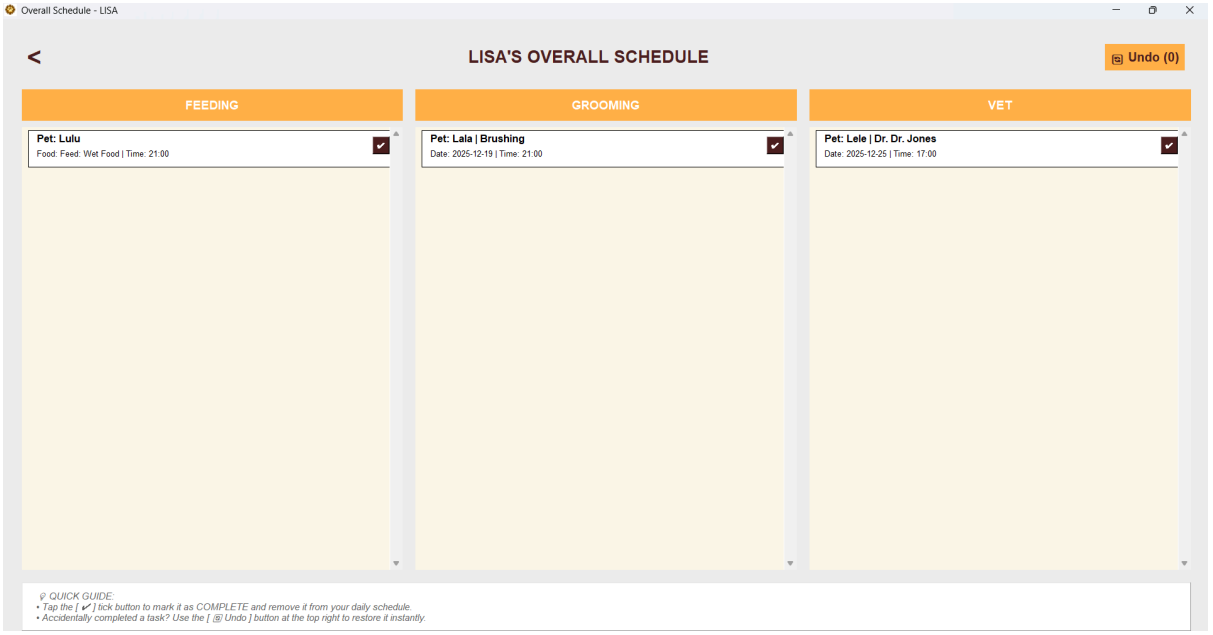
d)Lookback The Previous Remark
Users can also click the ‘Remarks’ to see the remark they had written.

e)Edit Record
-Users can also click the edit to update the appointment by clicking the “Edit” button and edit the input on the right part of the page to update the schedule .

f)Delete Record
- Users can also delete the record that they want to delete by clicking the “Delete” button to delete.

g)Back Button
-Users can click on the “<” button on the left corner to go back to the mainpage.

6.Overall Schedule



a)Overall schedule display at the three boxes
Users can check out the schedule for the current day on this page.

b)Tick Button
-Users can click on the tick button to confirm that the task has been done. After that , the task will be discarded.

c)Undo
-Users can click the undo button on the right corner of the overall schedule page to undo the changes.

d)Back Button
-Users can click on the “<” button on the left corner to go back to the mainpage.

🐾Test Cases

Login Page Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
LP-01	Login with valid username and valid password	Username = existing username in capital letters, Password = correct 6-digit number, click login button	Users login successfully and Main Page opens	Users login successfully and Main Page opens
LP-02	Login with incorrect password	Username = existing username, Password = wrong 6-digit number, click login button	Error message box displayed: “Incorrect password.”	Error message box displayed: “Incorrect password.”
LP-03	Login with non-existent username	Username = not in credentials.json, Password = any value, click login button	Warning message box displayed: “This username does not exist. Please Sign Up.”	Warning message box displayed: “This username does not exist. Please Sign Up.”
LP-04	Register new user with valid inputs	Username = all capital letter, Password = any value in 6-digits, click sign up button	Account is saved to credentials.json and success message displayed: "Account created successfully! You can now Login."	Account is saved to credentials.json and success message displayed: "Account created successfully! You can now Login."
LP-05	Username not in uppercase	Username = lowercase or mixed case Password = valid, click sign up button	Warning message shown: “Username must be ALL CAPITAL LETTER”	Warning message shown: “Username must be ALL CAPITAL LETTER”
LP-06	Register user with password	Username = all capital letter	Warning message shown:	Warning message shown: “password

	not in 6-digits	Password = more or less than 6-digits	“password must be 6 digits”	must be 6 digits”
LP-07	Register user with non-numeric password	Username = all capital letters, Password = letters/symbols, click sign up button	Registration is rejected with validation warning	Registration is rejected with validation warning
LP-08	Register using an existing username	Username = existed username, Password = valid, click sign up button	Information message displayed: “This username is already taken.Try Login”	Information message displayed: “This username is already taken.Try Login”
LP-09	Real-time username validation	Type lowercase letters into username field	Warning shows below the username field: “Username must be ALL CAPITAL LETTERS”	Warning shows below the username field: “Username must be ALL CAPITAL LETTERS”
LP-10	Real-time password validation	Type non-digit or incorrect length password	Warning shows below the password field: “Password must be 6 digits”	Warning shows below the password field: “Password must be 6 digits”

Main Page Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
MP-01	Main Page loads after login	Valid login	Main page open successful	Main page open successful
MP-02	Open Feeding Tracker	Click on Feeding Tracker button	Feeding Tracker opens	Feeding Tracker opens
MP-03	Open Grooming	Click on Grooming button	Grooming opens	Grooming opens

MP-04	Open Vet	Click on Vet button	Vet opens	Vet opens
MP-05	Open Overall Schedule	Click on Overall Schedule button	Overall schedule opens	Overall schedule opens
MP-06	Log out	Click on Log out button	Show confirm message and went back to Login page	Show confirm message and went back to Login page

Feeding Tracker Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
FT-01	Add a new feeding record	Input the informations	The record is saved and displayed in the history list.	The record is saved and displayed in the record list.
FT-02	Add new record with empty pet name	Leave pet name field empty, click Add / Save button	Error message displayed: “Pet Name is required!” and record is not saved	Error message displayed: “Pet Name is required!” and record is not saved
FT-03	Age field input validation	Type letters or more than 2 digits in Age field	Invalid input is rejected; only up to 2 digits are accepted	Invalid input is rejected; only up to 2 digits are accepted
FT-04	Select feeding date using calendar	Click Date field and select a date from calendar	Selected date is shown in the date field and calendar closes	Selected date is shown in the date field and calendar closes
FT-05	Edit an existing feeding record	Select a record, click Edit Selected, modify details, click Add / Save button	Selected record is updated with new values in the record list	Selected record is updated with new values in the record list

FT-06	Edit without selecting a record	Click Edit Selected without selecting any record	Warning message displayed: “Please select a record to edit.”	Warning message displayed: “Please select a record to edit.”
FT-07	Delete a selected feeding record	Select a record, click Delete Selected, confirm deletion	Record is removed from list and data is updated in JSON file	Record is removed from list and data is updated in JSON file
FT-08	Delete without selecting a record	Click Delete Selected without selection	Warning message displayed asking user to select a record	Warning message displayed asking user to select a record
FT-09	Clear all feeding history	Click clear all button, confirm action	All records are removed from list and JSON file is cleared	All records are removed from list and JSON file is cleared
FT-10	Back button navigation	Click the “<” back button	Feeding Tracker window closes and user returns to main page	Feeding Tracker window closes and user returns to main page

Grooming Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
G-01	Submit grooming schedule with valid inputs	Enter Pet Name, Description, select future Date, Time, Task, click Submit	New grooming record is saved and displayed in the records list	New grooming record is saved and displayed in the records list
G-02	Submit grooming schedule without Pet Name	Leave Pet Name empty, click Submit	Warning message shown: “Enter a Pet Name”; record is not saved	Warning message shown: “Enter a Pet Name”; record is not saved

G-03	Prevent selecting past date	Try selecting a past date from DateEntry calendar	Past dates cannot be selected due to mindate restriction	Past dates cannot be selected due to mindate restriction
G-04	Prevent scheduling grooming in the past time	Select today's date with a time earlier than current time, click Submit	Error message shown: "You cannot book an appointment at the past!Please select a new time."Appointment is not saved	Error message shown: "You cannot book an appointment at the past!Please select a new time."Appointment is not saved
G-05	View grooming record details	Select a record from the records list	Record details are displayed in the details text area	Record details are displayed in the details text area
G-06	Update existing grooming schedule	Select a record, modify details, click Update Schedule	Selected record is updated and changes are reflected in the records list	Selected record is updated and changes are reflected in the records list
G-07	Delete a selected grooming record	Select a record, click Delete, confirm deletion	Record is removed from the list and deleted from the file	Record is removed from the list and deleted from the file
G-08	Delete without selecting a record	Click Delete without selecting any record	Warning message shown: "Please select a record."	Warning message shown: "Please select a record."
G-09	Back button navigation	Click the "<" back button	Grooming window closes and user returns to main page	Grooming window closes and user returns to main page

Vet Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
V-01	Add a new vet	Enter Pet	Appointment is	Appointment is

	appointment with valid input	Name, Pet Age, select Vet, Visit Date, Visit Time, Remark, click “Save Appointment” button	saved successfully. A new record appears in the record list. Input fields are cleared after saving	saved successfully. A new record appears in the record list. Input fields are cleared after saving
V-02	Add record with non-selected date	Fill in all fields except Visit Date, click “Save Appointment” button	Error message displayed: “Please select a Visit Date.” Appointment is not saved, users remain on the same page.	Error message displayed: “Please select a Visit Date.” Appointment is not saved, users remain on the same page.
V-03	Add record with empty remarks	Leave Remark empty, click “Save Appointment” button	Error message displayed: “Please fill in all fields.” Appointment is not saved	Error message displayed: “Please fill in all fields.” Appointment is not saved
V-04	Add record with past date	Select a past date, select any time, click on Save Appointment button	Error message displayed: You cannot book an appointment at the past!Please select a new time.” Appointment is not saved	Error message displayed: “You cannot book an appointment at the past!Please select a new time.” Appointment is not saved
V-05	Edit existing appointment	Click Edit button on an existing appointment, modify details, click “Update Appointment” button	Appointment details is updated with new values, record list reflects updated information, save button text returns to “Save Appointment”	Appointment details is updated with new values, record list reflects updated information, save button text returns to “Save Appointment”
V-06	Delete existing appointment	Click the delete (🗑️) button, confirm deletion	Appointment is removed from record list,	Appointment is removed from record list,

			record file is updated, input form is cleared if the deleted record was being edited	record file is updated, input form is cleared if the deleted record was being edited
V-07	View appointment remarks	Click “Remarks” button on a record	Remarks pop-up window appears, displays correct remark text, Window, appears above the main page	Remarks pop-up window appears, displays correct remark text, Window, appears above the main page
V-08	Back button navigation	Click the “<” back button	Vet window closes and user returns to main page	Vet window closes and user returns to main page

Overall Schedule Test Cases

Test cases ID	Test Description	Input	Expected Result	Actual Result
OS-01	UI behavior when there are no tasks	Open Overall Schedule with no tasks in one or more categories.	Message “No Feeding Tasks!”, “No Grooming Tasks!”, or “No Vet Tasks!” is displayed, Application does not crash or freeze	Message “No Feeding Tasks!”, “No Grooming Tasks!”, or “No Vet Tasks!” is displayed, Application does not crash or freeze
OS-02	Verify that a task can be marked as completed.	Click the “✔” button on any task card.	Selected task is removed from the schedule, Task is recorded in undo history, UI refreshes immediately	Selected task is removed from the schedule, Task is recorded in undo history, UI refreshes immediately
OS-03	Verify that the undo function restores the last completed task.	Complete a task ,click the 🔄 Undo button	Last completed task is restored to its original category, Undo	Last completed task is restored to its original category, Undo

			counter decreases by 1 UI updates correctly	counter decreases by 1 UI updates correctly
OS-04	Verify behavior when undo is clicked with no available undo actions.	Click 🔄 Undo when no task has been completed.	Information message displayed: “No actions to undo.” No changes occur in the schedule	Information message displayed: “No actions to undo.” No changes occur in the schedule
OS-05	Verify scrolling functionality when there are many tasks.	Add many tasks to one category and scroll the column.	Scrollbar appears, user can scroll smoothly to view all tasks, no task overlaps or UI glitches occur	Scrollbar appears, user can scroll smoothly to view all tasks, no task overlaps or UI glitches occur
OS-06	Back button navigation	Click the “<” back button	Overall schedule window closes and user returns to main page	Overall schedule window closes and user returns to main page